ANSWERS

Exercise 1.1

1.(i) $A \times B = \{(2,1),(2,-4),(-2,1),(-2,-4),(3,1),(3,-4)\}$

$A \times A = \{(2,2),(2,-2),(2,3),(-2,2),(-2,-2),(-2,3),(3,2),(3,-2),(3,3)\}$

$B \times A = \{(1,2),(1,-2),(1,3),(-4,2),(-4,-2),(-4,3)\}$

(ii) $A \times B = \{(p,p)(p,q)(q,p)(q,q)\}$; $A \times A = \{(p,p),(p,q),(q,p),(q,q)\}$;

$B \times A = \{(p,p),(p,q),(q,p),(q,q)\}$

(iii) $A \times B = \{\ \}$; $A \times A = \{(m,m),(m,n),(n,m),(n,n)\}$; $B \times A = \{\ \}$

2. $A \times B = \{(1,2),(1,3),(1,5),(1,7),(2,2),(2,3),(2,5),(2,7),(3,2),(3,3),(3,5),(3,7)\}$

$B \times A = \{(2,1),(2,2),(2,3),(3,1),(3,2),(3,3),(5,1),(5,2),(5,3),(7,1),(7,2),(7,3)\}$

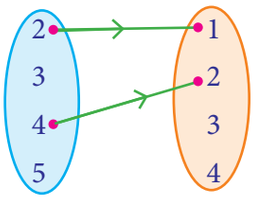
3. $A = \{3,4\}$ $B = \{-2,0,3\}$ 5. true

Exercise 1.2

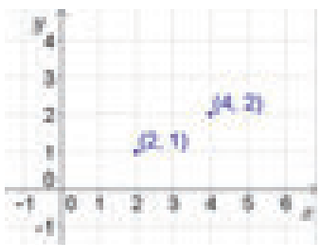
1.(i) Not a relation (ii) Not a relation (iii) Relation (iv) Not a relation

2. $\{1,2,3,4,5,6\}, \{1,4,9,16,25,36\}$ 3. $\{0,1,2,3,4,5\}, \{3,4,5,6,7,8\}$

4. (i)(a)

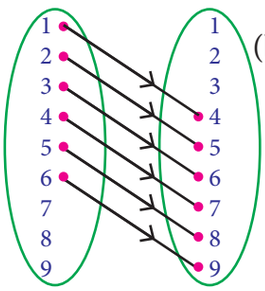


(b)

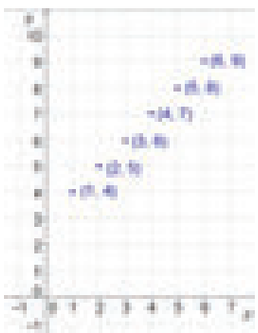


(c) $\{(2,1),(4,2)\}$

(ii)(a)

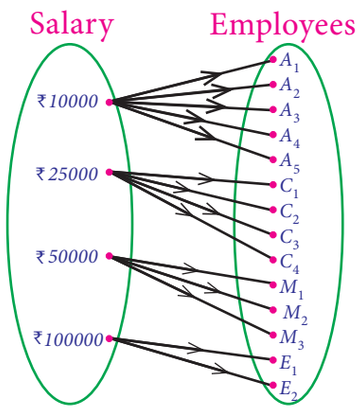


(b)



(c) $\{(1,4),(2,5),(3,6),(4,7),(5,8),(6,9)\}$

5. $\{(10000, A_1),(10000, A_2),(10000, A_3),(10000, A_4),$
$(10000, A_5),(25000, C_1),(25000, C_2),(25000, C_3),$
$(25000, C_4),(50000, M_1),(50000, M_2),(50000, M_3),$
$(100000, E_1),(100000, E_2)\}$



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